

Supplementary Material: Environment Sensitivity and Reproducibility of Post-BA SfM Benchmarks

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A. What this document supports

This supplementary substantiates four specific statements in the main paper; a reviewer following any of its pointers can go directly to the listed section.

- Protocol, “Track provenance”: “ ~ 0.209 is the reproducible level [for the released RESfM checkpoint] ... unchanged (0.217) under the authors’ exact pinned software stack” (Secs. B–D).
 - Protocol: “a stack-replication study (supplementary) shows the no-TTT arms are insensitive to it, while test-time fine-tuned means inherit solver-level nondeterminism” (Sec. E).
 - Table 2, † footnote: “TTT is run-to-run nondeterministic” (Sec. E).
 - Reading any post-BA mean in this literature: differences $\lesssim 0.05$ are within environment noise (Sec. F).
- In addition, Sec. G tabulates the per-scene results behind the main paper’s dataset-level numbers, Sec. H gives the complete training and evaluation configuration, and Sec. I provides the per-scene label-degradation breakdown behind the label-ceiling analysis.

B. The reproduction question

Evaluating the RESfM authors’ released checkpoint in our harness on the 36 MegaDepth test scenes yields a post-BA mean translation error of 0.209 vs. the published 0.169 (rotation 1.89° vs. 1.29°). Since our controlled comparisons hinge on baseline fidelity, we traced this discrepancy to its source. The finding is of independent interest: post-BA means on internet-photo benchmarks carry a practical environment-noise floor of $\mathcal{O}(0.05)$, produced by chaotic basin selection in robust bundle adjustment.

C. Eliminating the controllable factors

Data. Our MegaDepth npz files are byte-identical to the authors’ released archive, including the Group-2

Table 8. Released RESfM checkpoint, MegaDepth 36-scene post-BA means, under three conditions, vs. the published number.

Condition	Rot ($^\circ$)	Trans
Published (paper Table 1)	1.29	0.169
Our env (pip/OpenBLAS), 5-seed mean	1.82	0.203
Our env, seed 20	1.89	0.209
Authors’ exact stack (conda/MKL)	1.92	0.217

300-camera subsamples.

Randomness. Five evaluation seeds vary the pre-BA network output by ± 0.004 and the post-BA mean by ± 0.01 ; negligible.

Our re-implementation. Running the authors’ own evaluation code in our environment also yields ≈ 0.209 . Network forward pass. The pre-BA output (rotation/translation $11.88^\circ/1.730$, identical across seeds) is deterministic; the divergence therefore enters at the robust-BA stage. (Notably, the paper does not report pre-BA numbers; all its tables are post-robust-BA.)

Software stack. We rebuilt the authors’ exact pinned environment from their released environment.yaml: Python 3.8.20, MKL-backed numpy 1.24.3 (mkl=2023.1.0), pycolmap 3.10.0, pyceres 2.3, lightning 2.3.3. (The sole deviation is the CUDA build of PyTorch, forced by GPU-driver compatibility; the BA stage is CPU-only.) Re-evaluating all 36 scenes in this matched environment gives rotation/translation errors of $1.92^\circ/0.217$, statistically indistinguishable from our environment’s $1.89^\circ/0.209$, and not the published $1.29^\circ/0.169$.

D. Mechanism: chaotic basin selection in robust BA

Initialization perturbation is at the 10^{-10} level. We isolated the BLAS layer by running the identical DLT triangulation (the BA initialization) on identical inputs

under OpenBLAS and MKL, single-threaded: over 11,131 triangulated points the maximum relative difference is 4.8×10^{-10} (mean 5×10^{-14} ; NaN patterns identical).

Yet outcomes move by $\mathcal{O}(0.1)$. This sub-nanoscale perturbation changes the per-scene post-BA translation by up to 0.167, with 16/36 scenes moving by more than 0.01. Robust BA on ill-conditioned scenes amplifies last-bit differences into different convergence basins. The published gap has the same signature. Comparing our released-checkpoint evaluation to the paper’s Table 1 per scene: 22/36 scenes agree closely (many exactly), 13 scenes are better in our runs, and the entire mean gap is carried by four basin flips (scenes 0046, 0148, 0012, 5016; e.g. 0046: $0.043 \rightarrow 0.892$). Bidirectional, scene-concentrated flips are the signature of basin selection, not of a weaker checkpoint, which would shift errors in one direction rather than improve 13 of 36 scenes.

E. Stack replication and TTT run-to-run variance

Beyond the released checkpoint, we re-evaluated both from-scratch main-comparison checkpoints under the authors’ pinned stack (MKL numpy, pycolmap 3.10.0, pyceres 2.3), changing nothing else, and additionally repeated the TTT evaluation within our own stack. The supervised baseline is essentially unchanged across stacks: MegaDepth $0.344 \rightarrow 0.350$ mean, $0.070 \rightarrow 0.088$ median, only 7/36 scenes moving more than 0.05 (max ± 0.37 , bidirectional). This matches the released-checkpoint result above. The SS-RESfM weight+TTT arm, by contrast, varies regardless of stack: three evaluations of the same checkpoint (ours, a same-stack repeat, the authors’ stack) give MegaDepth means 0.316, 0.401, 0.453 (~ 18 – 20 /36 scenes shifting per repeat, up to ± 2.9 , largely the same scenes, medians 0.12–0.19). The attribution is therefore not the software stack but test-time fine-tuning itself: it runs an optimization at evaluation time, so the chaotic basin selection documented above re-enters the result on every run. Consequently the paper reports all results from the single declared pipeline, treats TTT means as carrying $\sim \pm 0.07$ run-to-run variance, and rests its environment- and run-independent claims on the deterministic no-TTT cells (which reproduce exactly on repeat).

F. Implications for reading the paper’s numbers

(1) The published number and ours are draws from a distribution over environments; no software matching can force a particular draw. (2) Differences

of $\lesssim 0.05$ in post-BA translation means on these benchmarks are within environment noise and should not be read as method differences, by any paper. (3) Consequently, meaningful comparisons must be environment-controlled: every number in the main paper (all arms, RESfM-from-scratch, and the released checkpoint) is produced in a single environment, and the out-of-distribution margins ($2\times$ and larger) sit far above this noise floor.

G. Per-scene results

Tables 9 and 10 report the per-scene post-BA translation and rotation errors and registered-camera fractions underlying the main paper’s dataset-level numbers (seed 20, the declared pipeline): the ten 1DSfM scenes and the 36 MegaDepth test scenes, for the three self-supervised arms (madweight, weight, weight+TTT) and the supervised RESfM baseline trained from scratch under the identical recipe. The per-scene view makes two aggregate statements concrete: on 1DSfM no single arm dominates, the best mechanism flips scene by scene (with *Ellis_Island* and *Tower_of_London* hard for every arm), consistent with the per-scene complementarity analysis in the main paper; on MegaDepth the four arms are close on most scenes and the dataset-level gap is dominated by a handful of high-error scenes. Table 11 completes the dataset-level record with rotation and registered-camera counts alongside translation. Rotation orders the arms consistently with translation on every dataset (including the supervised BlendedMVS failure, 36°). Registration is contamination-dependent: the supervised baseline registers the most cameras in distribution (85% vs. 78–80% on MegaDepth) and on clean Strecha, while under high contamination the label-free arms retain more cameras than supervision (38–43% vs. 37% on 1DSfM-hard) at lower error (madweight).

H. Implementation details

Table 12 lists the full training and evaluation configuration. Both methods share every setting except the base learning rate (10^{-4} self-supervised, 10^{-3} supervised, each selected on its own training objective) and the supervision source (COLMAP-derived labels vs. the adaptive percentile pseudo-labels).

I. Per-scene label degradation

Table 13 gives the per-scene data behind the main paper’s label-ceiling analysis: for each of the 18 scenes we measure the contamination level, the fraction of track observations that receive a COLMAP-derived label at

Table 9. Per-scene 1DSfM results. Post-BA translation and rotation error per scene (camera means; rotation in degrees), registered-camera fraction N_r (of the scene’s N_c ground-truth cameras), seed 20. Best translation per scene in bold.

Scene	N_c	madweight		weight		weight+TTT		RESfM	
		Trans/Rot	N_r	Trans/Rot	N_r	Trans/Rot	N_r	Trans/Rot	N_r
Alamo	577	1.77/2.45	81%	2.18/2.74	88%	1.67 /2.49	86%	1.73/2.42	87%
Ellis_Island	227	17.59/17.16	79%	15.90 /17.37	75%	17.11/17.74	78%	16.02/18.15	85%
Madrid_Metropolis	341	9.63/5.72	73%	4.37/2.84	88%	3.52/2.35	87%	3.38 /1.82	84%
Montreal_Notre_Dame	450	0.94/0.59	69%	11.35/0.55	72%	1.24/0.81	71%	0.93 /0.59	89%
Notre_Dame	553	0.68/3.33	92%	0.34/1.00	89%	0.33 /0.83	88%	0.49/1.08	89%
NYC_Library	332	2.94/2.32	85%	1.19 /1.16	79%	3.39/10.30	84%	6.28/12.99	86%
Piazza_del_Popolo	338	0.94 /0.43	64%	10.66/19.54	78%	17.05/20.46	75%	2.49/3.15	75%
Tower_of_London	472	76.34/35.61	67%	28.29 /23.30	71%	72.61/17.91	67%	67.95/49.66	63%
Vienna_Cathedral	836	7.30/7.42	64%	22.71/0.46	58%	33.57/26.80	78%	2.29 /1.09	59%
Yorkminster	437	5.79 /8.05	68%	30.87/14.95	85%	10.79/12.20	87%	8.77/6.88	71%

Table 10. Per-scene MegaDepth results. Post-BA translation and rotation error per scene (camera means; rotation in degrees), registered-camera fraction N_r , seed 20. Best translation per scene in bold. weight+TTT values are single draws of a run-to-run nondeterministic evaluation (Sec. E).

Scene	N_c	madweight		weight		weight+TTT		RESfM	
		Trans/Rot	N_r	Trans/Rot	N_r	Trans/Rot	N_r	Trans/Rot	N_r
0238	522	1.41/19.02	84%	0.62/8.23	26%	0.63/8.37	27%	0.12 /1.40	68%
0060	528	0.03/0.29	89%	0.02/0.20	90%	0.37/4.13	86%	0.01 /0.04	94%
0197	870	0.89/7.22	54%	0.38/3.54	39%	1.11/6.94	49%	0.11 /0.97	85%
0094	763	0.45/2.72	65%	0.16 /1.44	66%	0.17/1.34	57%	0.88/5.33	75%
0265	571	3.26/17.62	80%	2.34/5.78	84%	0.19 /0.60	96%	3.09/16.27	74%
0083	635	0.23/4.34	54%	1.56/17.70	80%	0.20/3.20	50%	0.10 /1.40	92%
0076	558	0.38/1.40	76%	0.23/0.61	93%	0.22/0.74	81%	0.02 /0.25	89%
0185	368	0.01/0.21	94%	0.04/0.24	91%	0.09/0.88	87%	0.01 /0.07	96%
0048	512	0.14/1.93	93%	0.02 /0.35	91%	0.03/0.43	88%	0.11/1.85	96%
0024	356	0.18/1.27	84%	0.17 /1.13	82%	1.13/4.78	81%	1.09/3.67	94%
0223	214	0.24/2.66	96%	0.23/1.69	93%	0.30/3.06	93%	0.08 /0.45	95%
5016	28	0.04/0.27	82%	0.02/0.08	89%	0.02/0.07	100%	0.01 /0.08	100%
0046	440	0.22 /2.88	96%	0.30/4.11	96%	0.25/3.88	95%	0.93/14.64	93%
0099	299	0.15/0.81	75%	3.52/2.77	72%	0.13 /1.58	71%	0.39/2.18	84%
1001	285	1.22/1.58	98%	0.73/1.49	98%	0.49 /1.22	98%	0.50/1.23	96%
0231	296	0.15/1.63	93%	0.07/0.64	87%	0.24/1.49	89%	0.02 /0.17	90%
0411	299	0.03/0.22	94%	0.32/2.99	82%	0.46/4.12	84%	0.02 /0.11	92%
0377	295	0.05/0.62	68%	0.26/2.76	75%	0.03 /0.56	82%	0.06/0.82	76%
0102	299	0.02/0.15	97%	0.01 /0.07	97%	0.01/0.08	97%	0.02/0.10	96%
0147	298	2.99/31.69	88%	1.21/15.28	79%	1.20 /15.27	76%	3.22/4.61	83%
0148	287	0.33/2.60	77%	0.11/2.11	70%	0.04/0.62	68%	0.04 /0.94	70%
0446	298	0.03 /0.40	98%	0.19/1.20	96%	0.19/1.04	98%	0.05/0.71	98%
0022	297	0.02/0.14	93%	0.03/0.22	92%	0.04/0.33	90%	0.02 /0.11	93%
0327	298	0.09/1.97	95%	0.02/0.22	96%	0.02/0.21	95%	0.01 /0.09	93%
0015	284	0.30/2.64	69%	0.56/3.87	69%	0.65/3.90	74%	0.30 /1.31	82%
0455	298	0.01 /0.20	95%	0.01/0.21	96%	1.60/0.46	97%	0.09/0.85	98%
0496	297	0.04/0.27	94%	0.04 /0.50	96%	0.09/1.28	97%	0.07/0.46	95%
1589	299	0.06/0.96	89%	0.05/0.28	88%	0.30/1.91	99%	0.01 /0.06	97%
0012	299	0.02 /0.23	94%	0.70/9.51	83%	0.22/2.76	88%	0.34/5.53	96%
0104	284	0.03/0.32	67%	0.18/2.28	68%	0.07/1.44	68%	0.02 /0.30	70%
0019	299	0.17/3.22	78%	0.28/6.55	85%	0.13/3.24	88%	0.01 /0.05	84%
0063	293	0.08/0.63	88%	0.07/0.72	91%	0.06 /0.65	91%	0.07/0.44	91%
0130	285	1.60/10.17	88%	0.16/1.87	65%	0.01 /0.33	65%	0.07/1.03	68%
0080	284	0.05 /1.10	46%	0.20/2.48	52%	0.07/0.70	52%	0.10/0.66	50%
0240	298	0.25 /2.23	92%	0.34/4.64	92%	0.46/5.32	91%	0.35/3.02	95%
0007	290	0.01 /0.07	37%	0.09/1.86	93%	0.14/3.28	90%	0.04/0.99	36%

162 all (coverage), and the precision and recall of those la-
 163 bels against ground truth. The aggregate trend in the

main paper is visible scene by scene: below $\sim 10\%$ con-
 tamination coverage is 0.92–1.00 and recall is 0.66–0.97

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Table 11. Dataset-level translation, rotation, and registered cameras (post-BA, seed 20) for the four main-comparison arms. Per-scene values are camera means; N_r/N_c is the fraction of ground-truth cameras surviving robust BA and the 5-pixel reprojection quality mask, summed over the dataset’s scenes.

	madweight	weight	weight+TTT	RESfM
Translation error (mean / median over scenes)				
MegaDepth	0.422/0.145	0.423/0.182	0.316/0.186	0.344/0.07
1DSfM	12.4/4.37	12.8/11.0	16.1/7.16	11.0/2.93
1DSfM-hard	12.2/8.83	21.9/13.1	26.4/17.7	16.2/13.1
Strecha	2.5/0.97	2/0.026	1.98/0.005	0.006/0.005
BlendedMVS	0.146/0.004	0.1/0.036	0.136/0.011	0.379/0.348
Olsson	2.57/1.15	2.65/1.22	7.74/0.18	8.47/0.17
Rotation error ($^\circ$, mean / median over scenes)				
MegaDepth	3.49/1.33	3.04/1.77	2.51/1.39	2.01/0.84
1DSfM	8.31/4.53	8.39/2.79	11.19/11.25	9.79/2.79
1DSfM-hard	14.40/7.39	25.58/17.70	31.97/12.83	22.70/15.86
Strecha	14.62/8.43	7.48/0.17	7.41/0.02	0.02/0.01
BlendedMVS	9.60/0.38	6.65/3.06	7.91/0.55	36.10/29.55
Olsson	13.57/4.37	13.23/5.41	10.49/2.27	12.34/3.35
Registered cameras N_r/N_c				
MegaDepth	80%	79%	78%	85%
1DSfM	74%	77%	80%	77%
1DSfM-hard	38%	39%	43%	37%
Strecha	91%	96%	98%	100%
BlendedMVS	95%	88%	95%	90%
Olsson	83%	90%	90%	88%

(mostly above 0.8), while above $\sim 15\%$ contamination recall never exceeds 0.50 on any scene (0.25–0.50) and coverage falls as low as 0.37: the supervision signal degrades exactly where outlier handling matters most.

Table 12. Training and evaluation configuration. All arms in the main paper use these settings; the only per-method difference is the base learning rate.

Setting	Value
Training (both methods)	
Epochs	20,000 (one epoch = one pass over all scenes)
Batch	4 scenes
LR schedule	milestones 10k, 11k, ..., 20k, $\gamma = 0.9$
Base LR	10^{-4} (SS-RESfM) / 10^{-3} (RESfM)
Loss weights	$\alpha_{\text{reproj}} = 1$, $\beta_{\text{cls}} = 1$
Infinity-point margin	10^{-4}
Checkpoint selection	best validation error, evaluated every 500 epochs
Seed	20 (21–22 / 21–24 for the seed studies)
Adaptive pseudo-labels (SS-RESfM only)	
Inlier / outlier percentile	20 / 80 (of per-observation reproj. error)
Threshold separation	scale-aware floor, $0.25 \times$ median error
Warm-up	10 epochs before pseudo-labeling starts
Evaluation	
Removal threshold τ	0.6 (0.8 for low-contamination single-camera sets, following RESfM)
Test-time fine-tuning	1,001 iterations, LR 5×10^{-3} , evaluated every 250 iterations
Bundle adjustment	pyceres/pycolmap robust BA, as RESfM

Table 13. Per-scene label-degradation results. For each scene: contamination (fraction of track observations that are GT outliers), COLMAP-label coverage (fraction of observations receiving a label), and precision/recall of the COLMAP-derived outlier labels against GT, sorted by contamination. Recall collapses above $\sim 15\%$ contamination.

Dataset	Scene	Contam. (%)	Coverage	Prec.	Recall
BlendedMVS	58c4bb4f4a...	0.5	1.00	0.86	0.94
Strecha	Herz-Jesu-P8	0.7	1.00	0.51	0.66
Strecha	fountain-P11	0.7	1.00	0.69	0.76
Strecha	Herz-Jesu-P25	2.6	0.99	0.72	0.82
Strecha	entry-P10	2.7	0.99	0.88	0.88
BlendedMVS	5be883a4f9...	4.0	0.98	0.90	0.84
BlendedMVS	5b6e716d67...	6.1	0.92	0.89	0.92
BlendedMVS	5bfd0f32ec...	7.1	0.97	0.97	0.97
BlendedMVS	5b950c7160...	9.2	0.96	0.92	0.94
MegaDepth	0007	11.2	0.73	0.84	0.47
1DSfM	Alamo	26.6	0.76	0.81	0.25
1DSfM	Madrid_Metropolis	28.8	0.67	0.85	0.28
MegaDepth	0041	31.4	0.49	0.80	0.42
MegaDepth	0060	39.0	0.42	0.53	0.45
1DSfM	Piazza_del_Popolo	41.9	0.76	0.90	0.30
MegaDepth	0047	54.9	0.37	0.83	0.50
1DSfM	Ellis_Island	57.9	0.79	0.89	0.37
1DSfM-hard	Gendarmenmarkt	59.8	0.50	0.89	0.29